

Assignment 1 - Data Transformation Strategy - Ma. Isabel Ortiz Naranjo

The objective of this assignment is to design and implement a data transformation process using Power Query in Power BI. The goal is to integrate multiple data sources into a single consolidated dataset that matches the structure provided in the Output file.

Data Sources

The solution uses six different data sources:

- Orders (TXT)
- Order Lines (TXT)
- Customers (Excel)
- Countries (CSV)
- Products (JSON)
- Age Groups (TXT)

Each source contributes different information required to build the final analytical dataset.

Transformation Process

The first step was importing all files into Power Query and validating their structure. Data types were standardized to ensure consistency across all datasets. The Orders file provided the order identifier, order date, and customer identifier. The Order Lines file contained detailed information about each purchased product, including units and unit price. The Customers file provided demographic information such as customer name, email, gender, date of birth, and country code. Customer age was calculated from the date of birth. The Age Groups file was used as a reference to classify customers into the categories Underage, Adult, and Old. Missing values were categorized as Unknown. The Countries dataset was used to enrich customer records with additional geographic information, including country, region, sub-region, and intermediate region. The Products dataset was imported from JSON format and expanded to include product names and product families.

Data Integration

After cleaning and transforming each dataset individually, the tables were merged using their corresponding keys:

- Customer ID
- Order ID
- Product ID
- Country Code

A new calculated field called **Amount** was created by multiplying the number of units by the unit price.

Final Result

The final output is a consolidated dataset that combines customer information, geographic attributes, order details, product information, and sales metrics into a single table. This approach improves data quality, simplifies reporting, and provides a structured dataset ready for analysis and visualization in Power BI.

Conclusion

The transformation process successfully integrated multiple data formats into a unified analytical model. By applying data cleaning, enrichment, calculated fields, and table relationships, the final dataset provides a reliable foundation for business analysis and reporting.